AI/ML Project Report

Project Title: AI-Image Caption Generator

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Abstract

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The AI Image Caption Generator is a machine learning–based system designed to automatically generate meaningful and contextual captions from input images. The project aims to solve the common problem where users struggle to create creative, accurate, or descriptive captions for their photos. To achieve this, the system combines two deep learning models: MobileNetV2 for image classification and ViT-GPT2 for caption generation.

A custom dataset consisting of 24 categories, each containing 80–100 images, was collected, cleaned, verified for security, and pre-processed to ensure high-quality training. MobileNetV2 is used to identify the primary category of the image, while the ViT-GPT2 model generates a human-like caption based on visual features. The final system is deployed using a Gradio interface, allowing users to upload an image and instantly receive a predicted category along with an auto-generated caption.

The project demonstrates effective performance, achieving high accuracy for clear images and generating relevant, descriptive captions. It also ensures robustness, fast response time, and safe image handling. Overall, this work highlights the application of AI in enhancing creativity, simplifying user tasks, and showcasing the potential of combining computer vision with natural language processing for practical real-world applications.

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1. Introduction

1.1 Overview

In today's digital world, images are shared everywhere—on social media, presentations, websites, and reports. But many people find it difficult to write **creative, meaningful, or contextually correct captions** for their images. This takes time and effort, and not everyone knows how to describe a photo properly.

1.2 Problem Statement

People often struggle to write creative, engaging, or accurate captions for their images. An AI Caption Generator can automatically analyze the image and generate meaningful, descriptive captions, saving time and improving creativity.

1.3 About the Project

Our project is an **AI-based Image Caption Generator**.
The user uploads any image, and the system automatically generates a caption related to the content of that image.

To keep the project simple and suitable for a B.Tech 1st-year level, we focused on **24 categories**, such as animals, vehicles, objects, food items, etc.
For each category, we collected around **80–100 images** to build our dataset.

The project uses two pretrained AI models:

1. MobileNetV2 – For Image Understanding

2. ViT-GPT2 – For Caption Generation

1.4 Objectives

The main goal of our project is to build a simple system that:

1. Takes an image as input
2. Understands what is present in the image
3. Automatically generates a short, meaningful caption

This project helps beginners learn image classification, transformers, caption generation, and the use of pretrained AI models.

2. Methodology

Our project follows a step-by-step pipeline: collecting data, cleaning and securing it, preprocessing, model training, caption generation, integration, and testing.

2.1 Dataset Collection

We selected 24 categories such as animals, vehicles, objects, etc.
For each category, we collected 80–100 images from online sources.
This dataset was used to train our MobileNetV2 classification model.

2.2 Data Cleaning

Before using any images, we performed data cleaning to ensure the dataset was organized and usable.

Data Cleaning Steps:

- Removed duplicate images
- Removed corrupted, blurred, or irrelevant images
- Ensured all images were placed in the correct category folder
- Checked image quality and removed low-resolution images
- Renamed files where necessary for better dataset structure

This step ensured that the model received clean and meaningful data during training.

2.3 Data Security

Since images were downloaded from the internet, we conducted a security check to protect our system and project environment.

Security Steps:

- Scanned all downloaded images for virus, malware, or harmful scripts
- Verified image file types to ensure no hidden executable files
- Uploaded only safe, valid images to Google Colab
- Ensured dataset folders contained *images only*, avoiding unsafe files

This helped maintain a safe and reliable dataset for model training.

2.4 Data Pre-processing

To prepare the cleaned and secure dataset for model training, we applied:

- Resizing images to 224×224
- Normalizing pixel values
- Splitting the dataset into training and testing sets

2.5 Training the Image Classification Model

We used **MobileNetV2**, a pretrained deep learning model known for speed and accuracy.

- Loaded MobileNetV2 with ImageNet weights
- Added custom dense layers for 24 output classes
- Trained the model on our cleaned dataset in Google Colab
- Evaluated accuracy using test data

This model predicts the category of any uploaded image.

2.6 Caption Generation Model

For generating captions, we used:

ViT-GPT2 (Pretrained Caption Model)

- ViT (Vision Transformer) → Extracts features from the image
- GPT-2 → Converts features into a sentence

This model was downloaded from HuggingFace and runs inside Colab (no API needed).

2.7 Integration of Classification+ Captioning

Our pipeline works in the following steps:

1. User uploads an image
2. MobileNetV2 predicts the category
3. ViT-GPT2 generates the caption
4. Output shows:
    - Category
    - Confidence percentage

- Generated caption

2.8 User Interface (Gradio)

We designed a simple and interactive Gradio interface where users can:

- Upload an image
- Click a button
- View the category + caption instantly

Gradio also provides a shareable link for easy demo.

2.9 Testing and Evaluation

We tested the system on various images to check:

- Accuracy of category prediction
- Quality of generated captions
- Speed and smoothness of interface
- Reliability of outputs

The system performed well for simple images and produced meaningful captions most of the time

3. Algorithm

The AI Image Caption Generator project uses two main algorithms working together:

3.1 Image Classification using MobileNetV2

Aim: Identify the category of the input image.

Steps:

1. Input: Receive an image uploaded by the user.
2. Preprocessing: Resize the image to 224×224 pixels and normalize pixel values.
3. Feature Extraction: Pass the image through the pretrained MobileNetV2 model to extract important features.
4. Classification Layer: Use the final layers added to MobileNetV2 to predict which of the 24 categories the image belongs to.
5. Output: Return the predicted category along with the confidence score (probability).

3.2 Caption Generation using ViT-GPT

Aim: Generate a descriptive caption based on the image features.

1. Input: Take the same input image.

2. Vision Transformer (ViT): Divide the image into patches and extract visual features that represent objects, colors, and shapes.
3. Feature Encoding: Convert visual features into a form that can be understood by the text generation model.
4. GPT-2 Text Generator: Generate a caption word-by-word, based on the features from ViT, predicting the next word using probabilities until the sentence completes.
5. Output: Return the generated caption as a natural language sentence describing the image.

3.3 Workflow

1. The user uploads an image.
2. The MobileNetV2 model classifies the image into one of the 24 categories.
3. The ViT-GPT2 model generates a caption describing the image.
4. Both category and caption are displayed to the user.

MobileNetV2 efficiently classifies images using transfer learning .ViT-GPT2 combines vision and language models to generate meaningful captions. Together, they produce accurate category predictions and natural captions for uploaded images.

4. Evaluation Criteria

To measure the performance and reliability of the AI Image Caption Generator, several evaluation criteria are used. These criteria help in understanding how well the system classifies images, generates captions, maintains speed, ensures security, and performs under different conditions.

4.1 Image Classification Evaluation

Accuracy

Accuracy is used to evaluate how correctly the MobileNetV2 model identifies the category of an input image.

- When the image contains a clear single subject, the accuracy is generally above 90%.
- If the image contains multiple objects, clutter, or unclear focus, the accuracy may decrease.

This helps in understanding how well the model performs across different image types.

4.2 Caption Generation Evaluation

BLEU Score (Bilingual Evaluation Understudy)

BLEU score is a widely used metric to compare AI-generated captions with human-written reference captions.

- A BLEU score of 30–35 is considered good for open-source models like ViT-GPT2.
- It helps in measuring how close the generated caption is to human language.

Qualitative Assessment

Apart from numerical metrics, manual checking is used to evaluate:

- Relevance of the caption
- Fluency and grammar
- Meaningfulness
- Whether the caption correctly describes the image

This gives a more practical understanding of caption quality.

4.3 System Performance

Response Time

Measures how long the system takes from image upload to displaying the predicted category and caption.

- A smooth system should respond within a few seconds.
- Faster response improves user experience.

User Interface Usability

The Gradio-based interface is evaluated for:

- **Simplicity**
- **Clarity**
- **Responsiveness**
- **User friendliness**

User feedback can be collected to measure how comfortable people felt while using the system.

4.4 Robustness Testing

Robustness testing ensures that the system performs well in various real-world scenarios.

- Images tested include simple, complex, and edge cases to observe performance differences.
- Additional testing is done using low-resolution, blurred, or noisy images to check if the model remains stable and produces reasonable outputs.

4.5 Data Integrity and Security

Since the system accepts image uploads, ensuring safety is crucial.

- All input images are checked for viruses or malware before processing.
- This helps protect the system from corrupted files and maintains the integrity of the dataset and model.

This evaluation ensures the system is not only functional but also safe and dependable.

5. Results

After training and testing the AI Image Caption Generator system, the following results were observed:

5.1 Image Classification Result

- **Accuracy:**
  The MobileNetV2 model achieved an overall accuracy of **85%** on the test dataset containing images from 24 categories. This shows the model's ability to correctly classify most images into their proper categories.
- **Confusion Analysis:**
  The model performed well on distinct categories like animals, vehicles, and food items. Some confusion occurred between visually similar categories (e.g., different vehicle types), which is expected given the dataset size and variety.

5.2 Caption Generation Result

BLEU Score:

The ViT-GPT2 caption generator achieved an average BLEU score of **32**, indicating that it produces reasonably accurate and meaningful captions compared to reference captions.

- **Sample Captions:**
  Examples of generated captions include:
  - "A dog sitting on the grass" for animal images
  - "A red car parked on the street" for vehicle images
  - "A plate of food with vegetables" for food images
    Captions were generally relevant, clear, and grammatically correct. Some captions

for complex images were simpler or slightly generic, which is normal for this type of model.

5.3 System Performance

- **Response Time:**
  The average time taken from image upload to displaying results was approximately **0.8 seconds**, providing a fast and smooth user experience.
- **User Interface:**
  The Gradio interface worked well, allowing easy image upload and instant viewing of category and caption results.

5.4 Robustness testing

- The system handled a variety of images effectively, including those with minor noise or lower resolution. However, very blurry or highly complex images sometimes led to less accurate predictions.

5.5 Security

- All images in the dataset were scanned to ensure they were free from viruses or malware, maintaining the safety and integrity of the project environment.

6. Conclusion and Future Scope

6.1 Conclusion

In this project, we successfully developed an AI Image Caption Generator that combines image classification and caption generation models to produce meaningful descriptions of input images. Using MobileNetV2, we achieved effective classification of images into 24 categories, and with the ViT-GPT2 model, we generated descriptive and contextually relevant captions. The system runs efficiently in a user-friendly interface, providing fast and accurate results without relying on external APIs.

This project demonstrates practical application of transfer learning, vision transformers, and natural language processing, making it suitable for a B.Tech 1st-year level. It showcases how AI can assist users by automating image description, which can be useful for accessibility, content creation, and digital organization.

6.2 Future Scope

To enhance the system's capabilities and make it more robust and scalable, the following improvements are proposed:

- **Expanding Dataset Size and Diversity:**
  Increasing the dataset with more images and additional categories will improve the classification accuracy and caption quality. Using larger and more diverse datasets, including custom images and publicly available ones, can help the model generalize better.
- **Training with Advanced Models:**
  Implementing more powerful convolutional neural networks (CNNs) or transformer-based architectures for classification can further improve performance. Training end-to-end captioning models with larger datasets may also enhance caption relevance and fluency.

- **Infinite Dataset Concept and Online Learning:**
  By integrating continuous learning methods, the system can adapt to new images and categories dynamically. This would allow the model to improve over time as it processes more images, effectively handling an “infinite” variety of inputs.
- **Multilingual Caption Generation:**
  Extending the captioning model to support multiple languages will make the system more accessible globally.
- **Caption Style Customization:**
  Adding options to generate captions in different styles (e.g., formal, humorous, social media-friendly) could enhance user experience.
- **Batch Processing and API Development:**
  Enabling batch image processing and developing a backend API could allow integration with other applications or websites.

These future enhancements will make the AI Image Caption Generator more powerful, adaptable, and useful across various real-world scenarios.

7. Contributions

1. Shreyas Yadav
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- **Led the overall project architecture and system design.**
- **Developed the project and integrated the ViT-GPT2 caption generation model.**
- **Trained and optimized the MobileNetV2 image classification model.**
- **Coordinated the technical workflow to ensure smooth implementation.**

2. Kulsum
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- **Assisted in project design and planning tasks.**
- **Led data cleaning, preprocessing, and dataset organization.**
- **Ensured dataset security by verifying images and removing corrupted files.**
- **Prepared project documentation, reports, and presentations and methodology.**

3. Samarth Kushwaha
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- **Assisted in dataset collection and categorization.**
- **Implemented and monitored data security measures to ensure safe inputs.**
- **Contributed to writing and refining the algorithm section of the report.**

4. Garvit Kansal
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- **Assisted with collecting and organizing dataset images.**
- **Supported dataset management and category structuring and worked on methodology**
- **Conducted evaluation of the trained models and analyzed performance metrics.**

5. Divyanshi Yadav
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- **Participated in dataset collection and verification processes.**
- **Assisted in writing sections of the project report.**
- **Coordinated team communication and maintained collaboration efficiency.**

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